

TRSI - Homework 1

Proposed problem

Usually, to obtain a good deep neural network model you need a lot of data for it to be trained on. In the context of rail scene semantic segmentation, there is not a lot of data to train such algorithms (the biggest dataset is RailSem19 containing 8500 pairs of image-masks, some of the annotations also not being that precise), so the motivation behind this research report would be to generate some data in the given scenario and to try to train a semantic segmentation algorithm to prove that some synthetic data would improve the performance of the current SOTA algorithms.

The proposed problem is a generational one, so we need some models that translate some kinda input into our desired output. Because translating from a plain text to both a realistic image and its segmentation mask would be a challenging task for one model and for the human as well (to encapsulate all the desired data into a given prompt), we propose a 2 step generation algorithm: one that translates structural prompts into segmentation masks (ex: an environment specific to a tramway, 10 people on the cross walk, 3 cars on the road and near another tram circulating in the opposite way), and one that takes as input some environmental prompt and a segmentation mask and generates a realistic image (ex. the tram should be near a mountain and there should be some snow on the rails). Later on, the model could be used without any input (so the output would be kinda random) if only train data is wanted to be generated.

Input data

At step 1, the model could take a structural prompt (described in the proposed model), but is optional, if no environmental prompt is given, the model would take only some default tokens. At step 2, the model could take an environmental prompt (described in the proposed model), but is optional, also, the model should take a 512x512 segmentation mask as control (to control the output), if none is given, the model generates the output with an empty mask (a black one).

Output data

At step 1, the model will output the generated segmentation mask, which would be a 512x512x3 picture, the significance of each label being described in the documentation of the model / application. At step 2, the model will output the generated realistic rail specific image, which would be a 512x512x3 picture.